

Geoinformation

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Chair of Cartography and Visual Analytics

WS 25/26

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L2: Introduction – Part 2

III. Different views of space and time

IV. Challenges of geoinformatics

III. Different views of space and time

Mathematical view

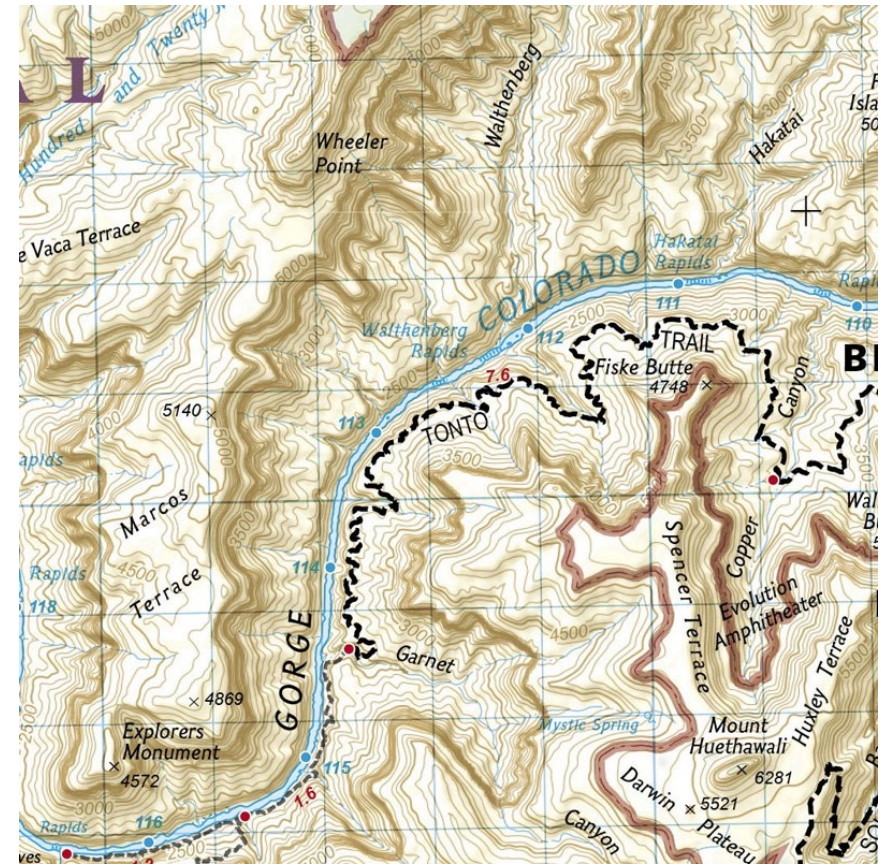
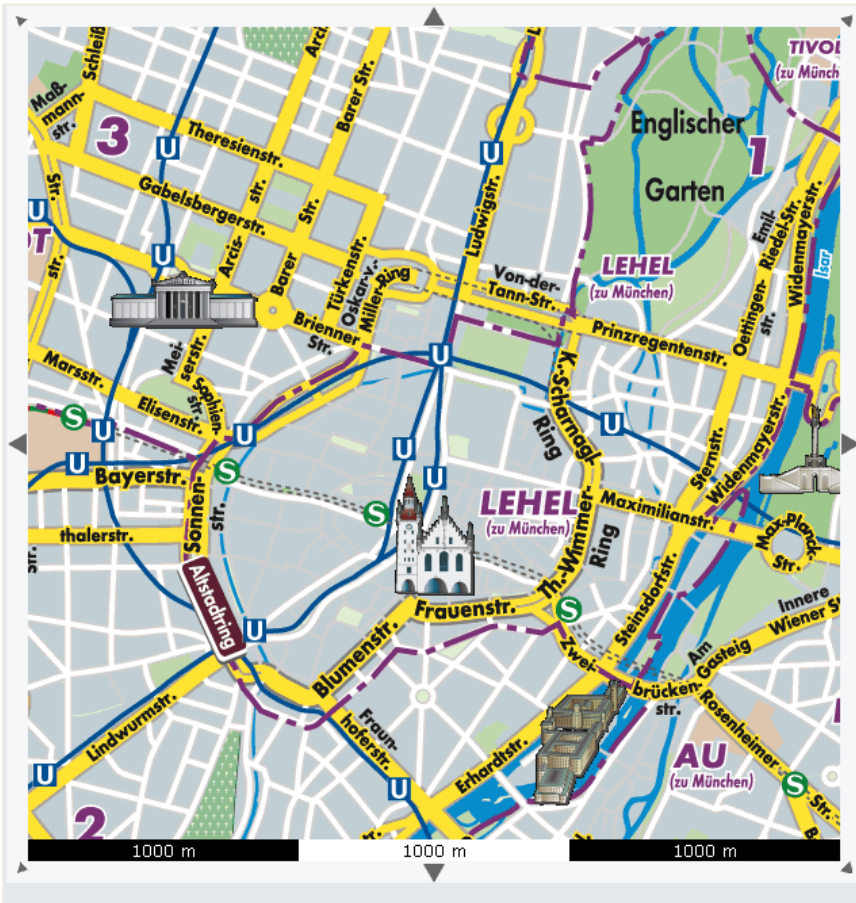
- Euclidean space (e.g. the Euclidean distance):

$$d_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

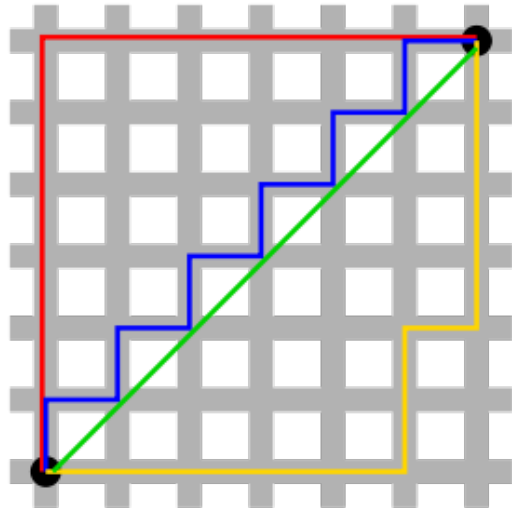
- Gridded space, i.e. 'Manhattan' or 'taxicab' metric:

$$d_{i,j} = |x_i - x_j| + |y_i - y_j|$$

- Topological space: invariant properties under some transformations, e.g. inside and outside, touching, connection.

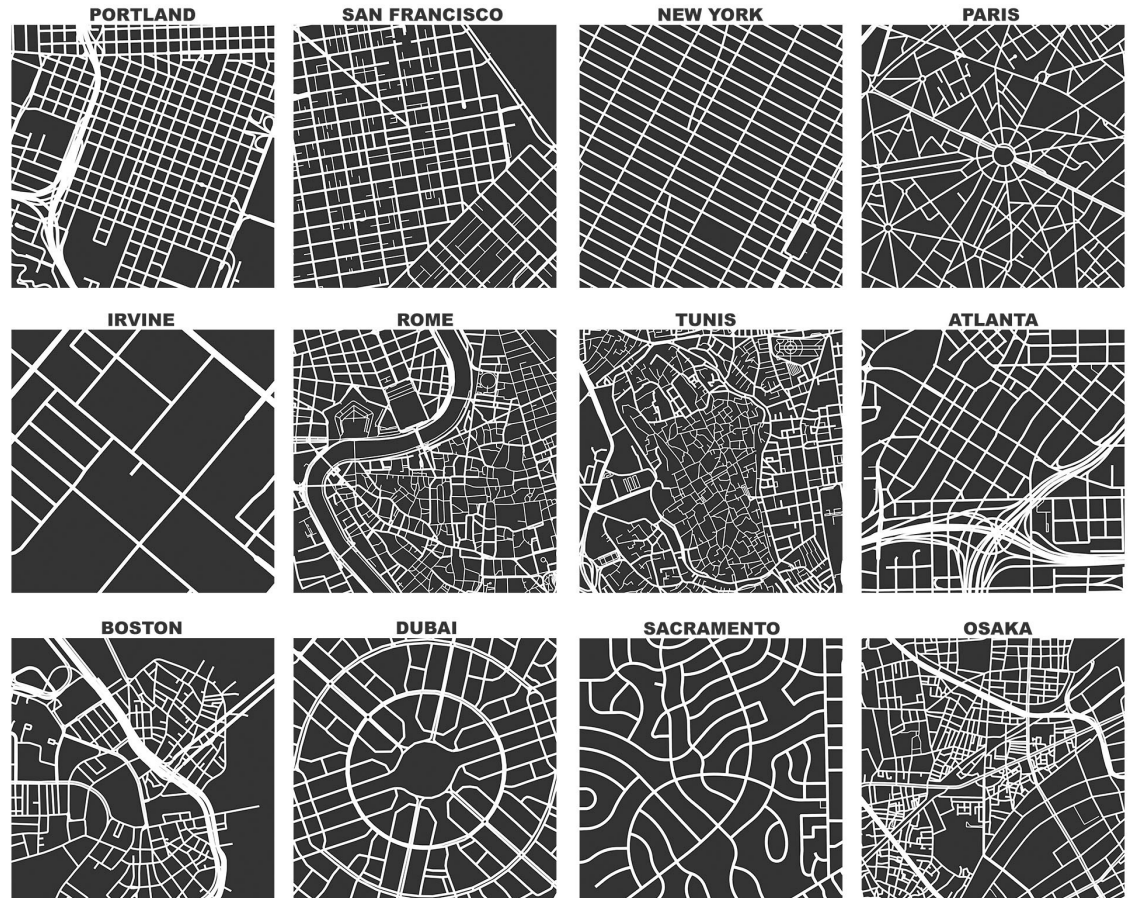


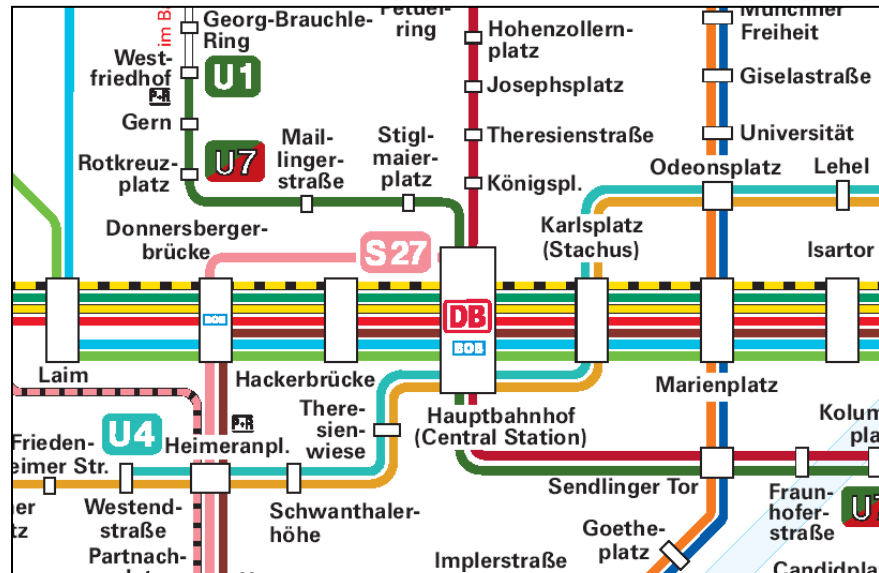
Euclidean



Gridded

Suitable for regular grid patterns

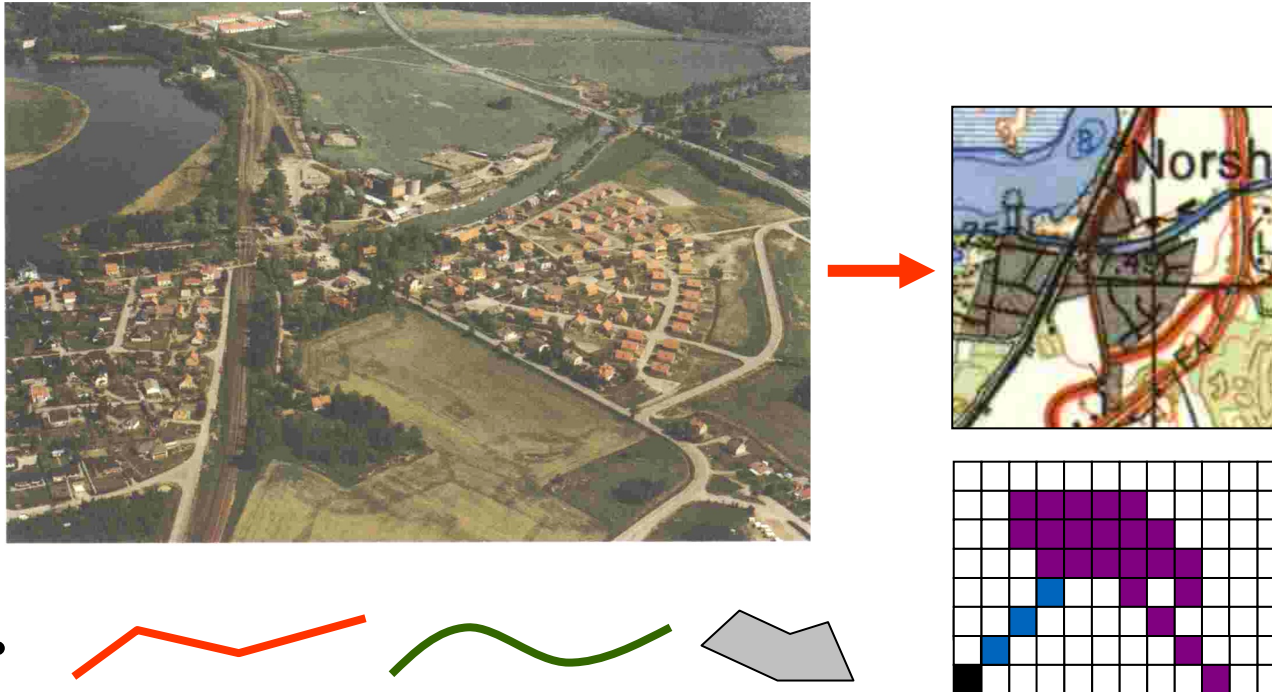


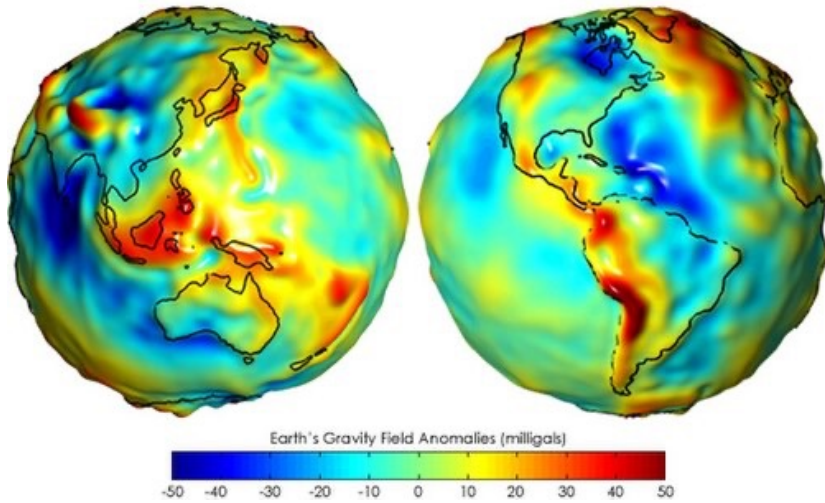
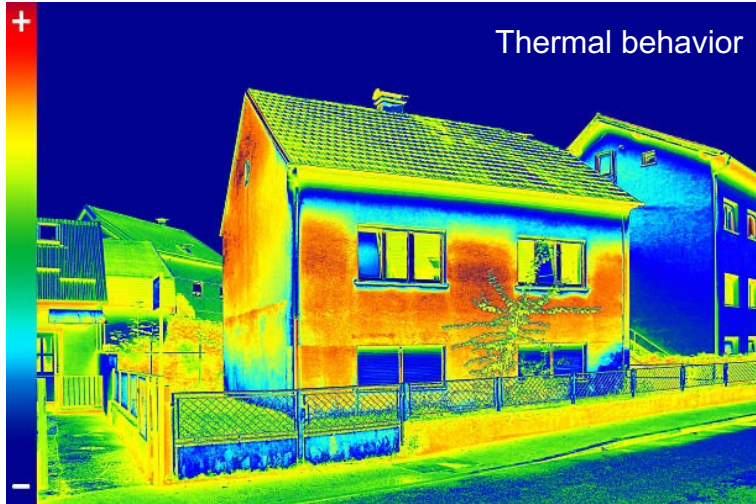


Topological

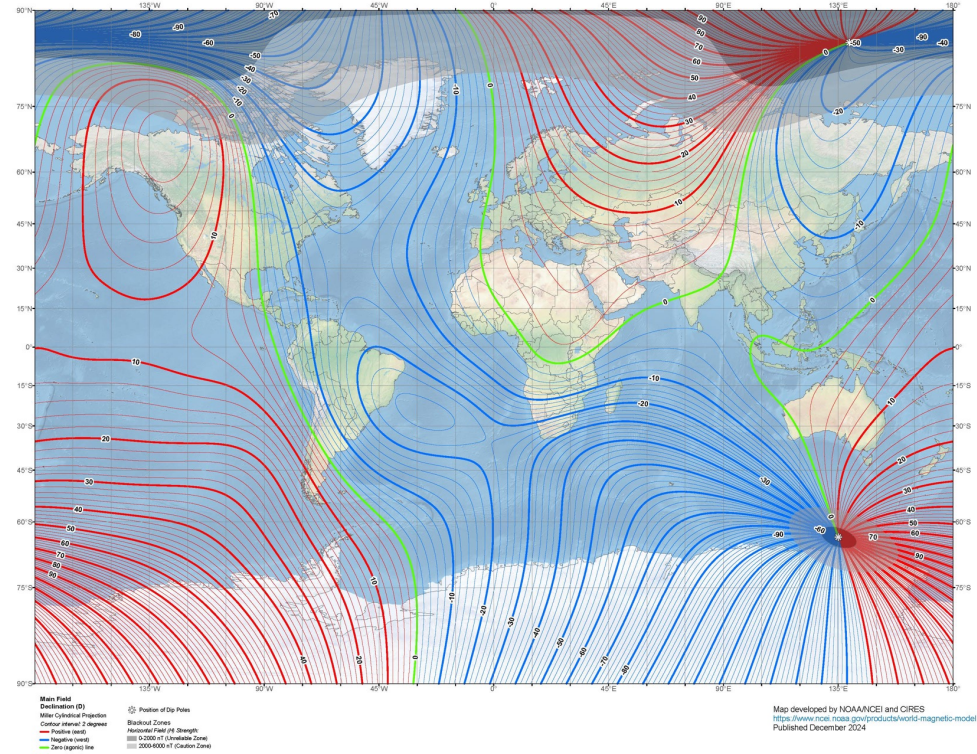
Physical view

Real-world objects can be measured and represented by geometric primitives, e.g. a church as a point, a road as a line, or a lake a polygon or a connect set of pixels.





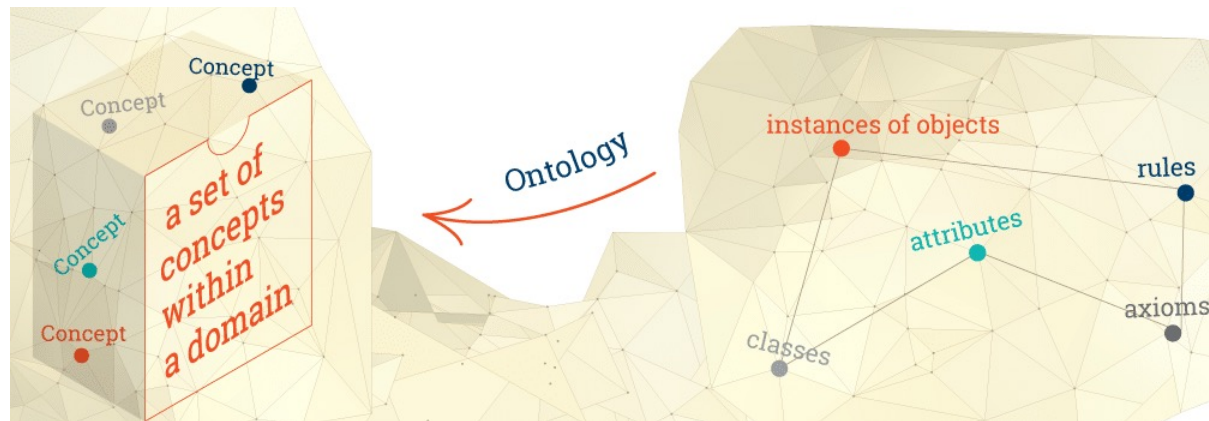
US/UK World Magnetic Model - Epoch 2025.0
Main Field Declination (D)



Invisible but measurable physical features

Philosophical view

The world is an all inclusive body of knowledge (ontology) and can be described using concepts, categories and relationships within a domain and across domains in space and time at different scales.

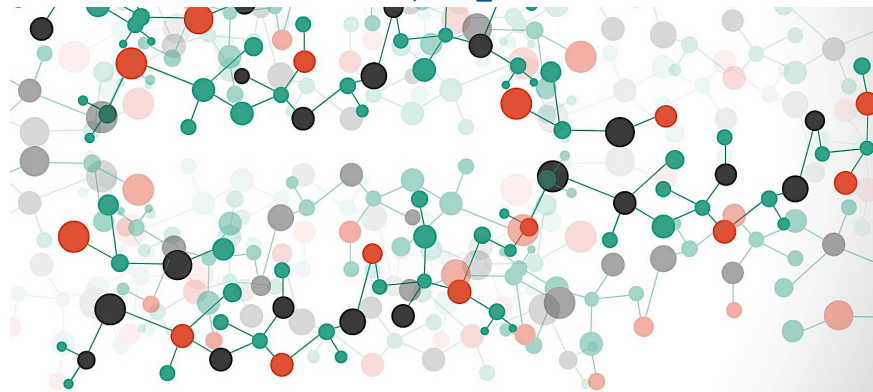


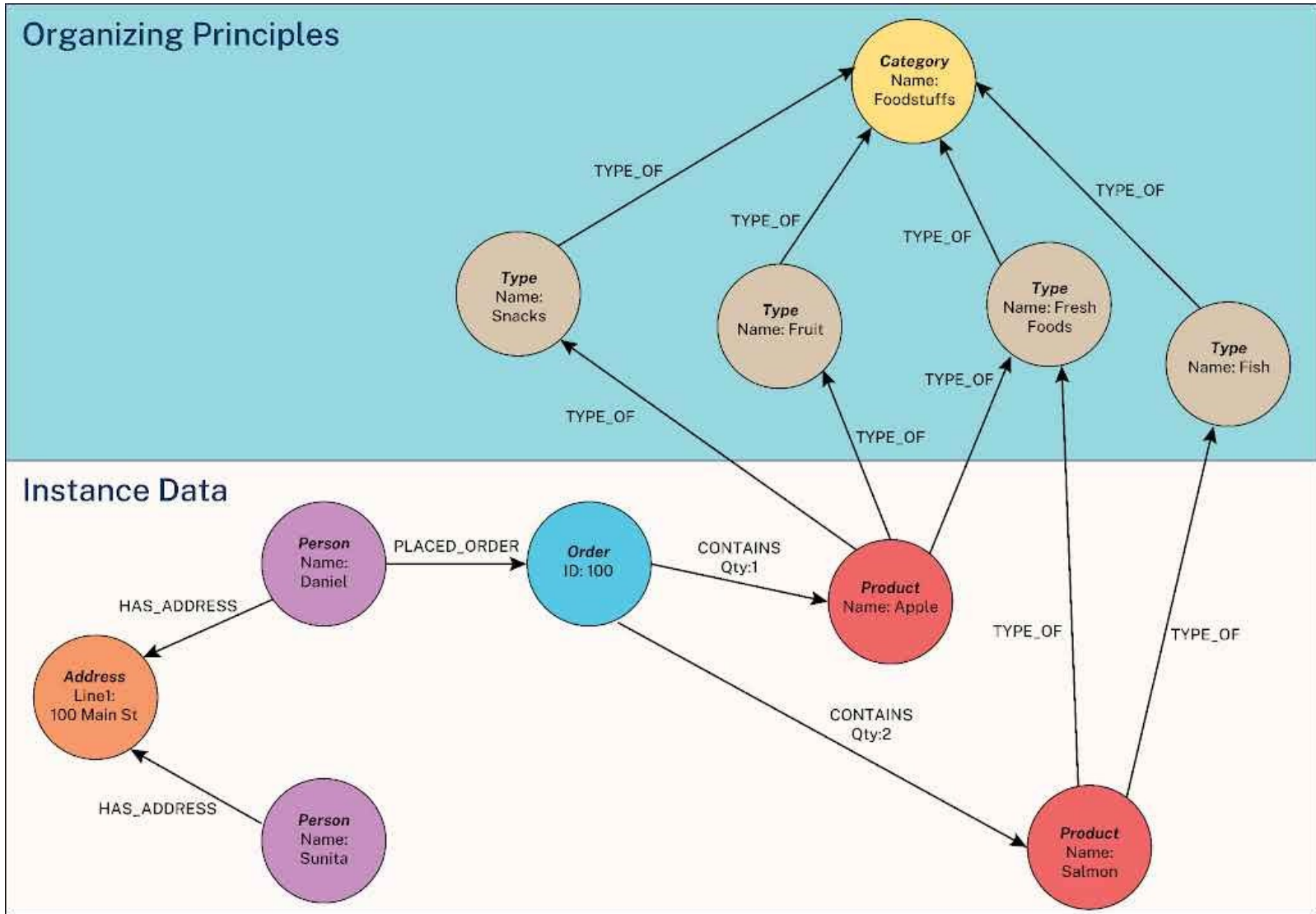
instantiation

abstraction

www.ontotext.com

Knowledge graph





An example

Create a knowledge graph

Fictive news 1 from Daily ABC, Oct 18, 2025

A recent analysis of satellite imagery has confirmed what local farmers have claimed for months: the White Nile River has shifted its course. Over the past two years, a combination of severe drought and a massive landslide 500 km upstream has caused the river to carve a new channel, moving ca. 2 km to the east.

The new river course has placed a 50 km² area of fertile farmland, previously part of South Sudan, on the eastern bank, which is sovereign territory of Ethiopia. The farmers who have worked this land for generations now find themselves in a different country without moving an inch. The foreign ministries of both nations have to resolve the situation, highlighting how dynamic processes can challenge static political borders.

Create a knowledge graph

Fictive news 2 from Daily ABC, Oct 19, 2025

The city council of Phoenix has passed a "Cool Roofs" mandate following the release of a startling heat map of the city. The map, created by researchers at Arizona State University, used satellite data to reveal a temperature difference of up to 8°C between the dense, paved city center and the surrounding desert and parklands — a typical Urban Heat Island effect. "It's not enough to know where things are, we have to now *why* they are that way," said Dr. Lena Reyes, the lead researcher. "

Under the new policy, all new commercial buildings and major renovations within the city's urban core must be constructed with light-colored, reflective roofing materials. The city is also offering subsidies to homeowners in these areas to repaint their roofs.

Create a knowledge graph

Fictive news 3 from Daily ABC, Oct 20, 2025

A massive construction project for a new residential complex in central Berlin was abruptly halted today after workers unearthed a 500-kg unexploded bomb from World War II. The discovery occurred at a site on the corner of Wilhelmstrasse and Niederkirchnerstrasse. The bomb was located using a combination of historical aerial photographs from 1945 and modern ground-penetrating radar. Experts from the German bomb disposal unit, have cordoned off a 1,000 m evacuation zone, affecting over 5,000 residents, and a segment of the U-Bahn subway line.

A spokesperson of the German bomb disposal unit stated, "This is a reminder that the past is buried just beneath our feet. Our work relies on overlaying historical maps with modern city plans to keep people safe." Disposal experts expect the defusing operation to take most of the day.

Cognitive and socio-cultural views

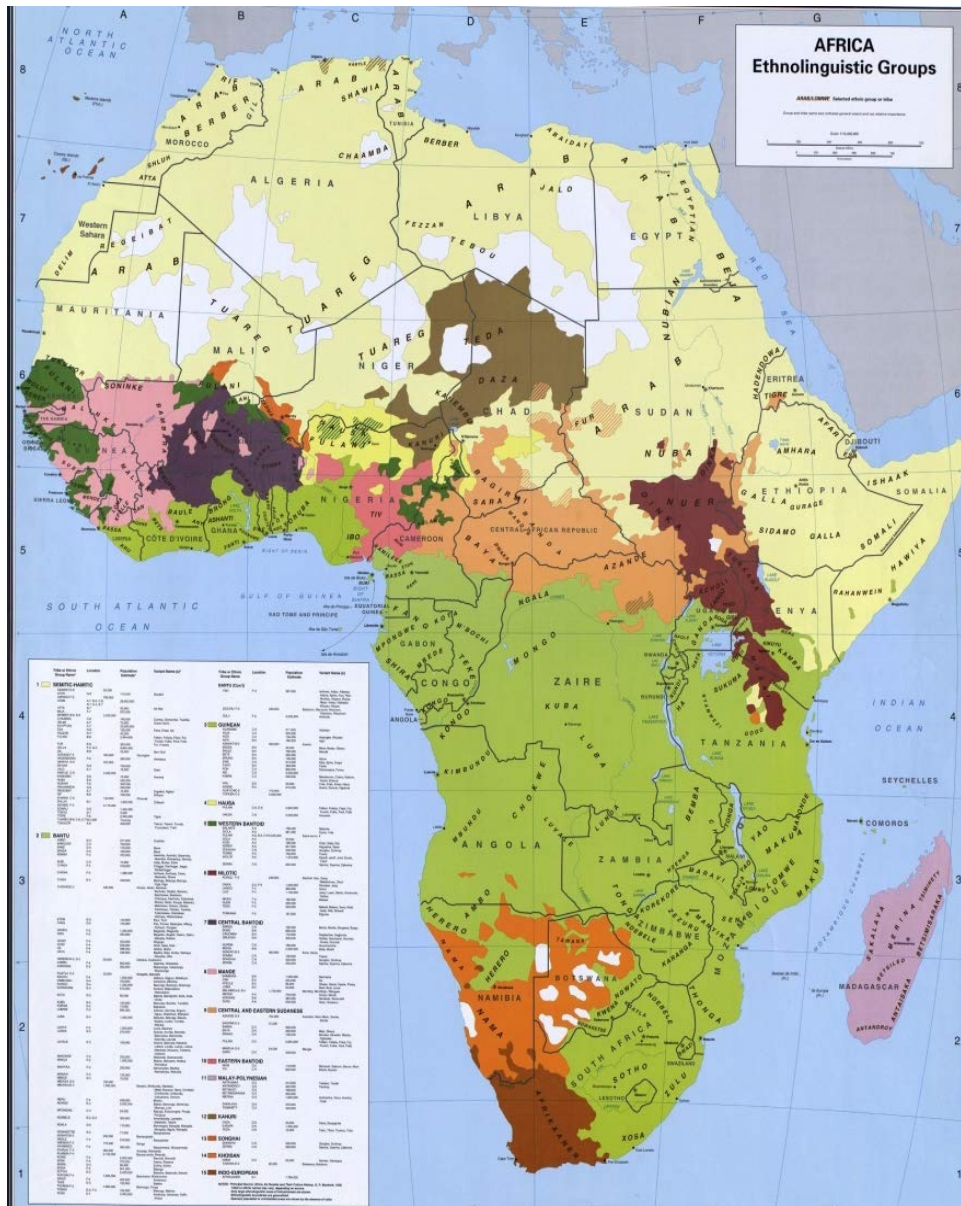
- There is no single reality that is the same for all. Different realities exist in different minds and different application domains.
- Cognitive factors often influence behaviour more than objective properties of the world may do.
- Social phenomena vary across geographical space with different spatial reference units.

The world of a child

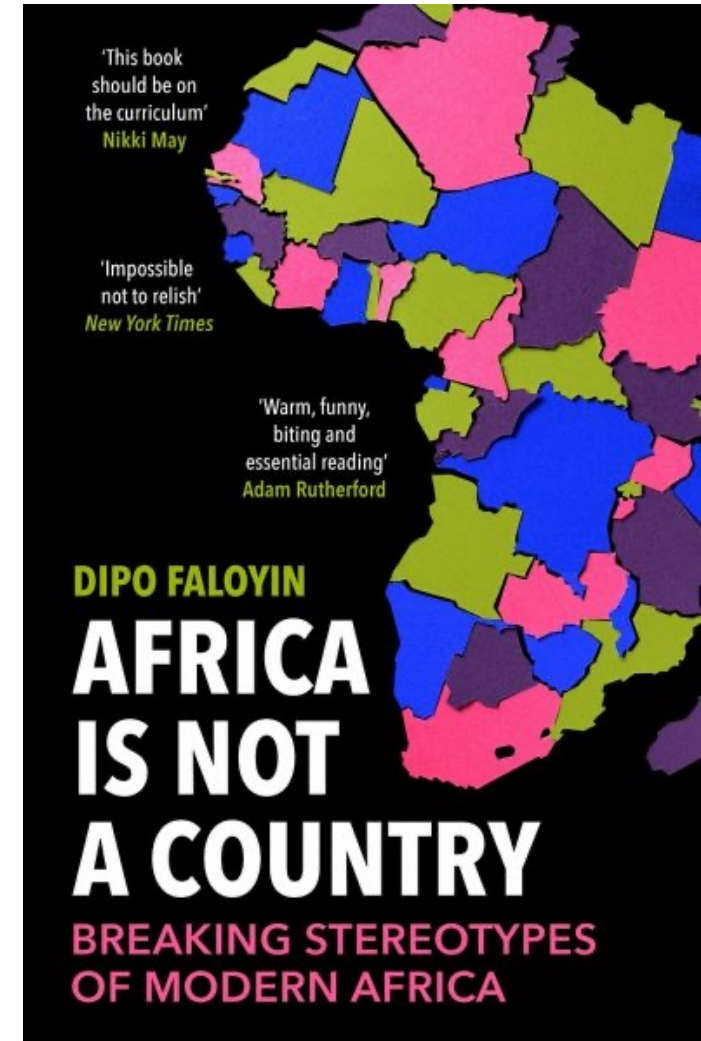


Bulgarien - Sofia - Aleksandar V. Cerovski - *The children draw and make a better world*

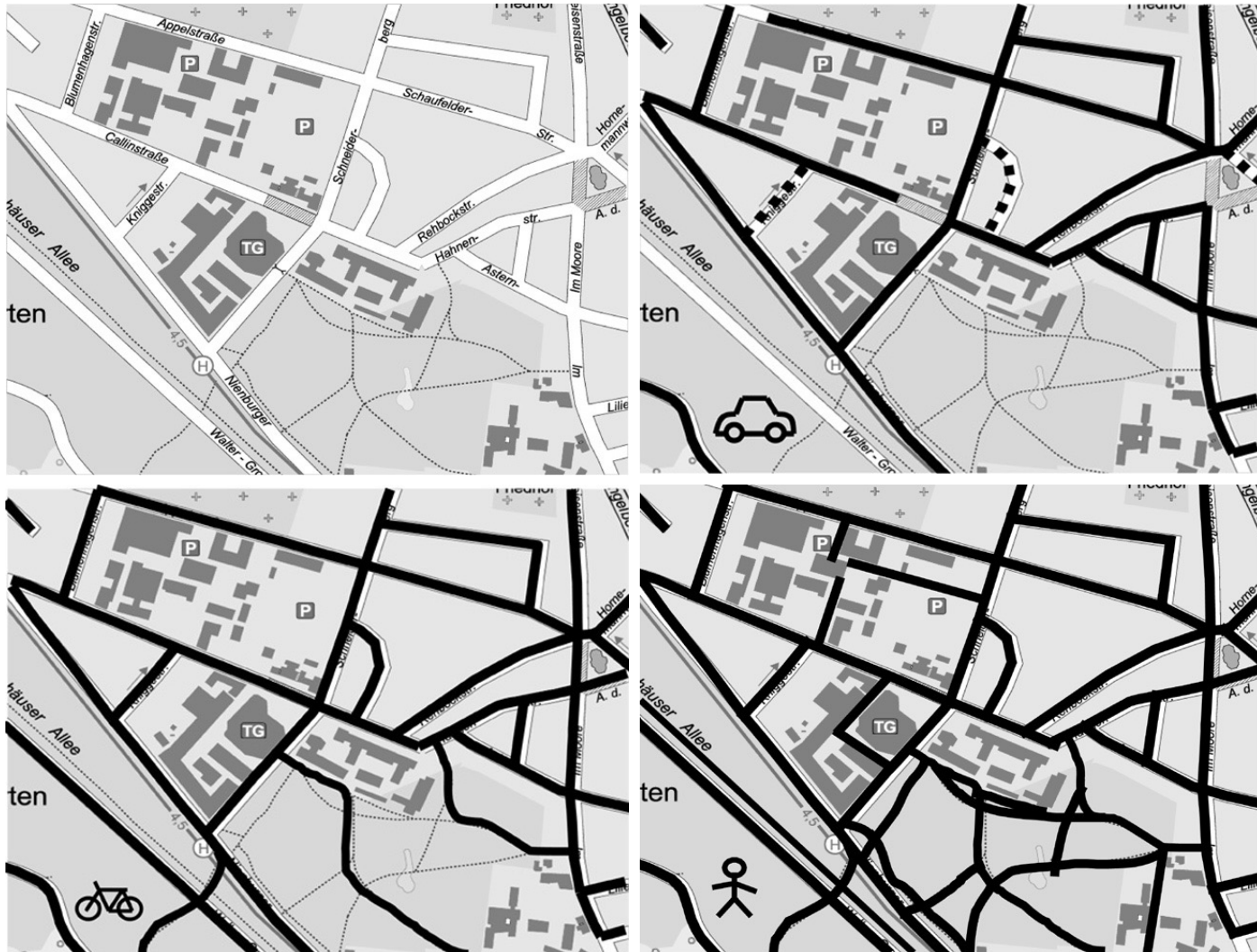




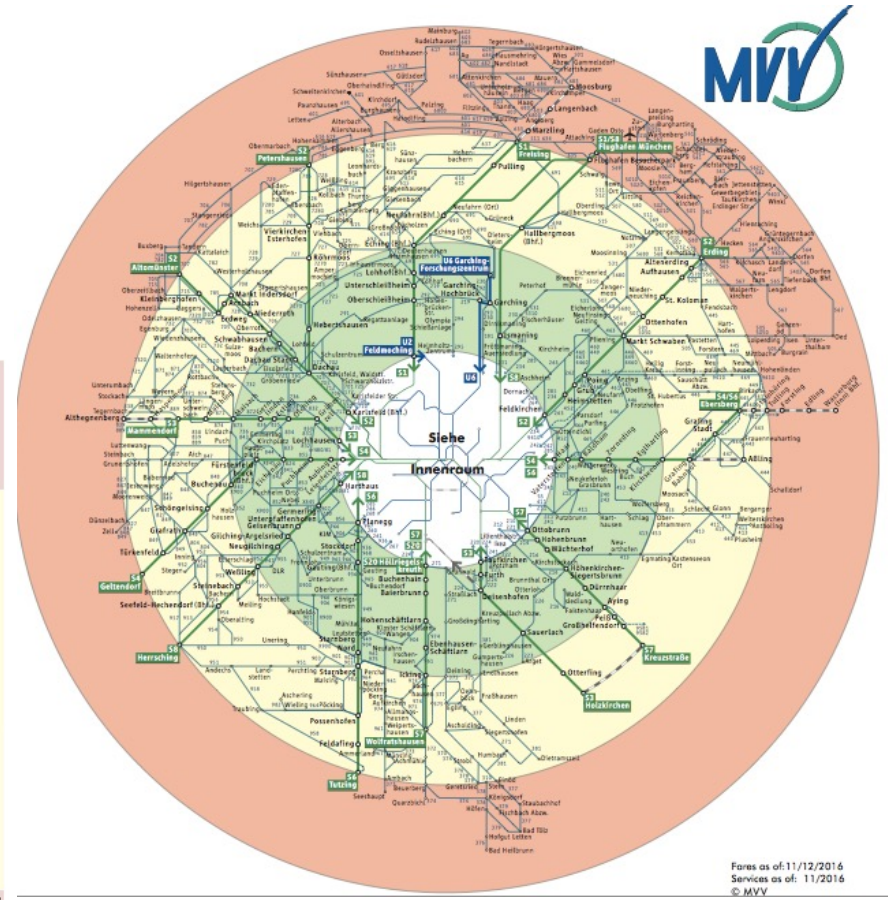
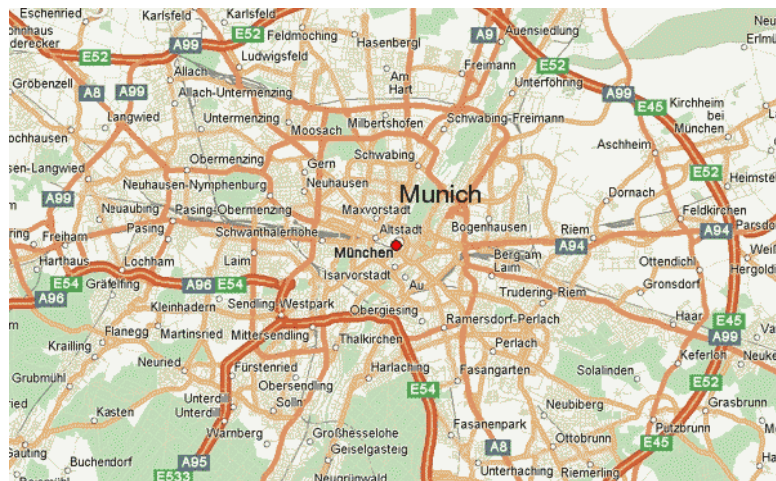
Amazon.de



2022



Different views of the same city



IV. Challenges of geoinformatics

Modeling and handling of time

- A lot of information is missing for measurements with different time stamps.

Modeling and handling of non-metric space

- It is difficult to compare and represent non-metric relationships such as proximity in terms of distance.

Modeling and handling of uncertain / biased properties

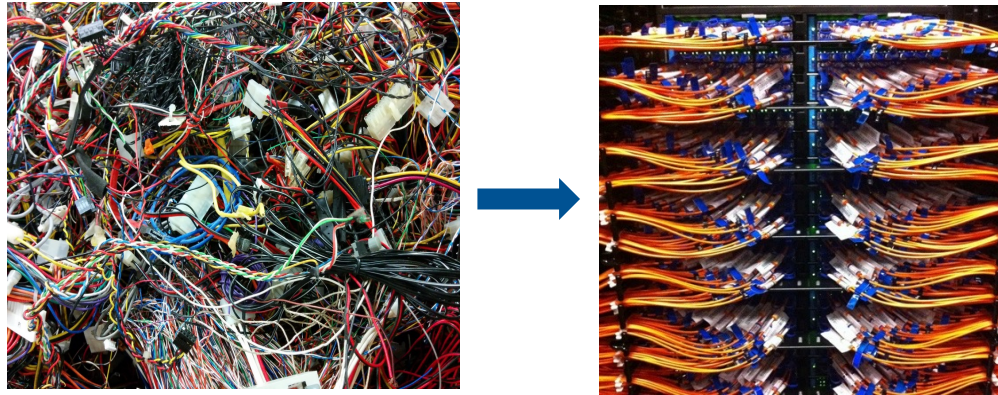
- The observed properties of and relationships between entities, fields and phenomena are uncertain and often biased.

Modeling and handling of multiple views of the world

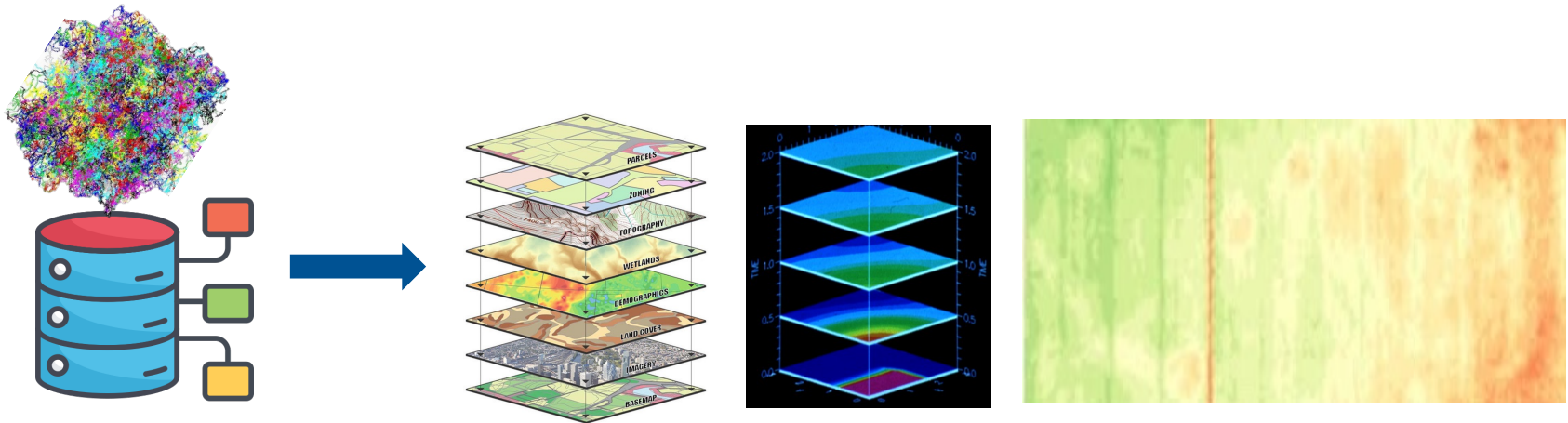
- It is difficult to transform multiple and sometimes conflicting insights into a uniform data model. There is no single correct view.

Promotion of open geodata

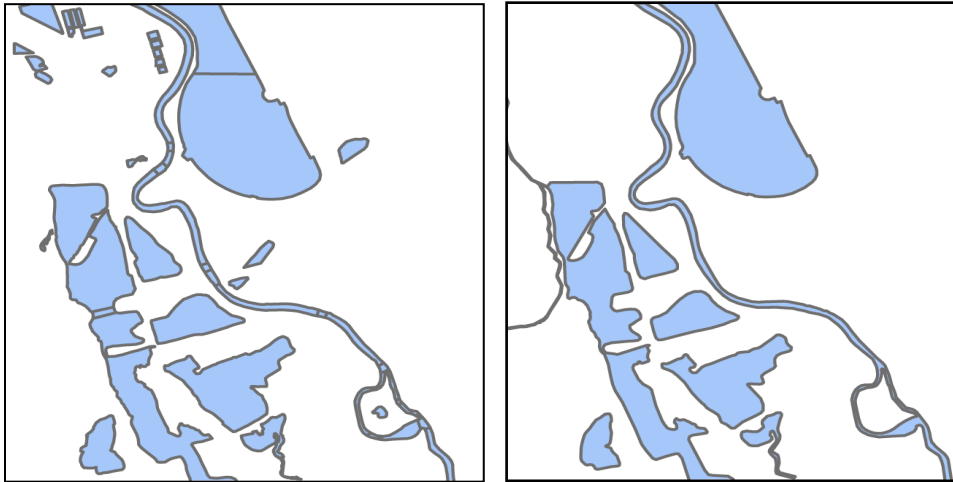
- The contribution and consumption of open geographic data are imbalanced.



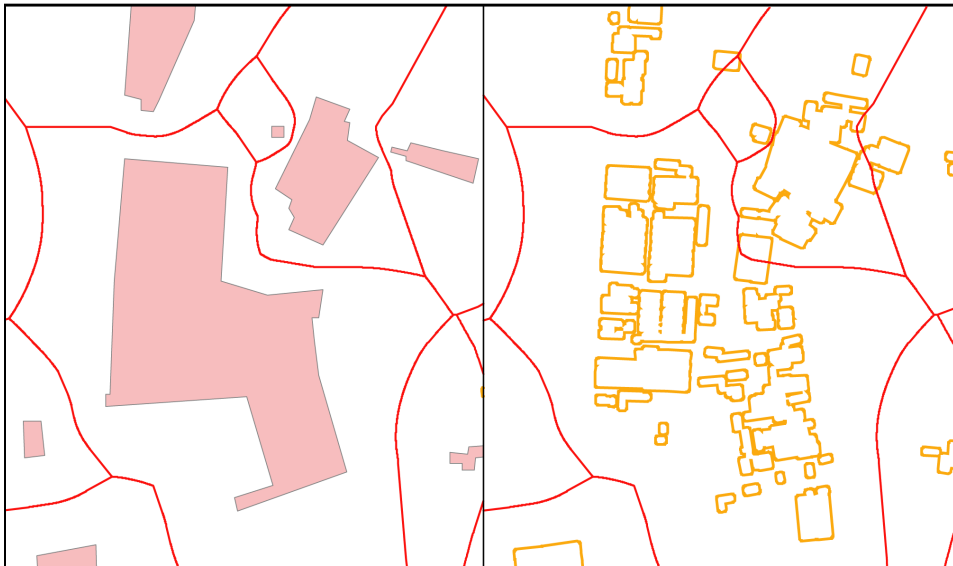
<https://towardsdatascience.com/the-importance-of-layered-thinking-in-data-engineering-a09f685edc71/>



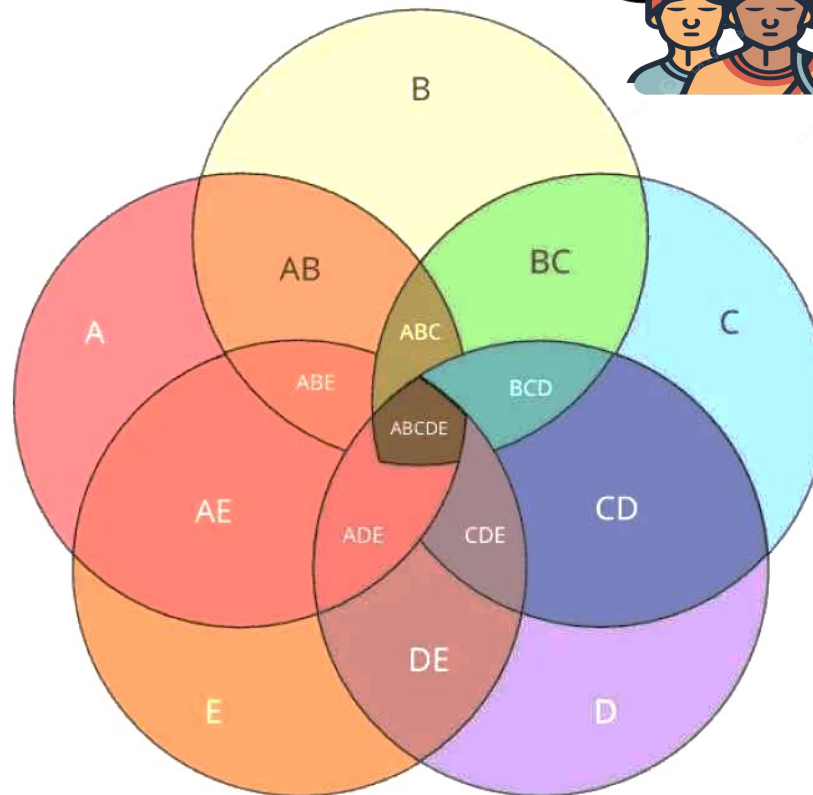
slicing and ordering



Datasets of the same region and scale



Datasets of the same region but different map scales



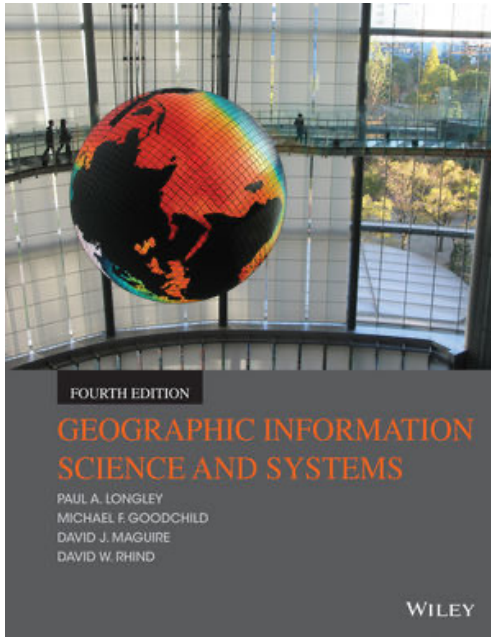
Shared facts may serve as a foundation for collaboration,
even when worldviews differ.



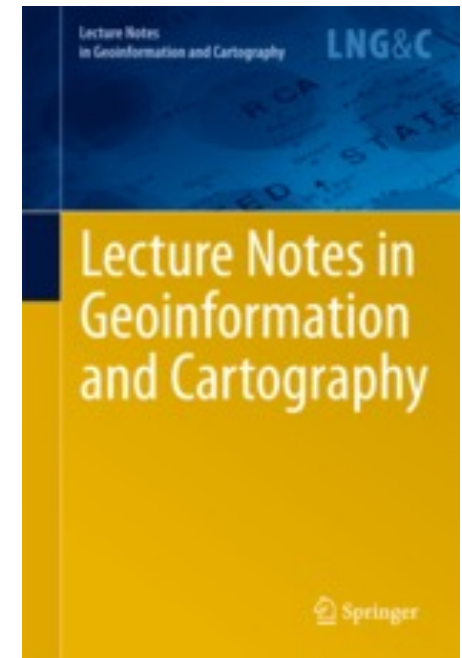
- It is people who turn locations into places and shape the livability of environments.
- It is difficult to capture and understand the in-situ knowledge and feelings of individuals.



Main references



- PA Longley, MF Goodchild, DJ Maguire & DW Rhind (2015):
Geographical Information Science and Systems



- Lecture Notes in Geoinformation and Cartography since 2006
Editors: W Cartwright, G Gartner, L Meng & MP Peterson



1. Explain Euclidean, Manhattan (taxicab) and topological space using application scenarios.
2. How do you understand the dimensions of geoinformation?
3. What makes the ontological view of space and time different from physical view?
4. Explain the main research challenges of geoinformatics using examples.
5. How would you handle multiply views of the world in a project?